

CHAPTER 14

CHEMICAL KINETICS

14.1 Factors Affecting Reaction Rates

14.2 Reaction Rates

14.3 Concentration & Rate (Rate Law)

14.4 Concentration & Time

14.5 Temperature & Rate

14.6 Reaction Mechanism



14.1 Factors Affecting Reaction Rates

1. Physical State of Reactants

rate increases with increasing number of collisions between reacting molecules.

2. Reactant Concentrations

rate mostly increases with increasing concentration.

3. Reaction Temperature

rate generally increases with temperature.

4. Presence of a Catalyst

catalysts usually increase reaction rates.



14.2 Reaction Rates

- Chemical Kinetics: deals with speeds (rates) of reactions. How fast a reaction will go?



Rate: change in concentration (Δ conc.) with change in time (Δt).

$$\text{Rate} = \frac{\Delta[\]}{\Delta t}$$

Rate of either: disappearance of reactants
or appearance of products



Reaction Rate:



$$\Delta[A] = [A]_f - [A]_i$$

$$\Delta[B] = [B]_f - [B]_i$$

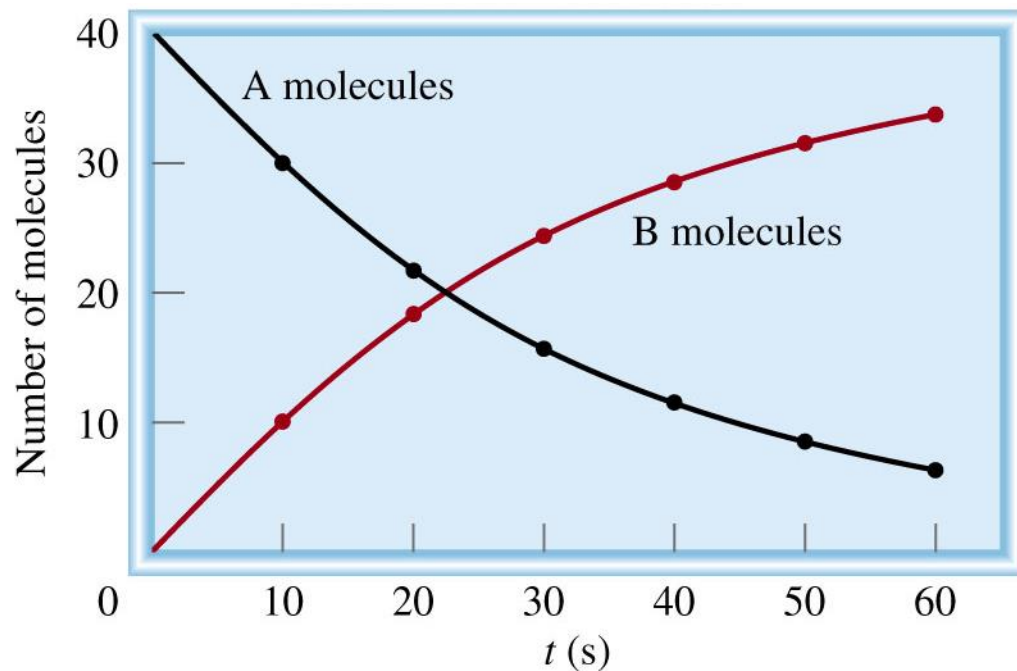
$$\text{rate} = \frac{-\Delta[A]}{\Delta t}$$

$$\text{rate} = \frac{\Delta[B]}{\Delta t}$$

$$\text{Units} = \frac{M}{s}$$

$$\Delta[A] = - \quad \text{decreasing}$$

$$\Delta[B] = + \quad \text{increasing}$$



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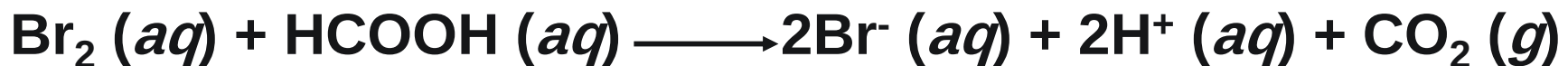
brown

colorless



time





Time (s)	[Br ₂] (M)
0.0	0.0120
50.0	0.0101
100.0	0.00846
150.0	0.00710
200.0	0.00596
250.0	0.00500
300.0	0.00420
350.0	0.00353
400.0	0.00296

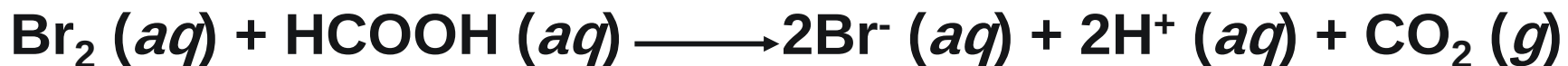
$$\text{average rate} = - \frac{0.00296 - 0.0101}{400 - 50.0}$$

$$= 2.04 \times 10^{-5} \text{ M.s}^{-1}$$

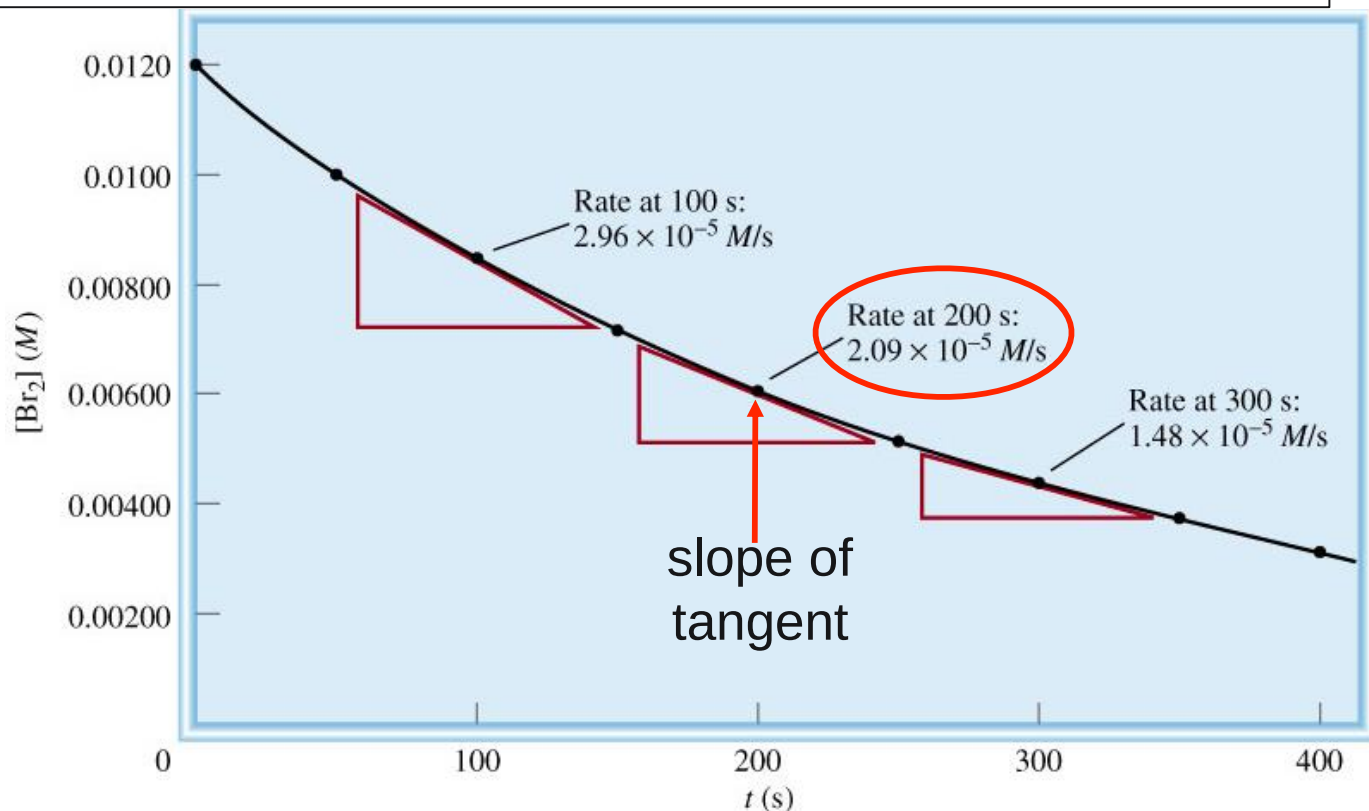
$$\text{average rate} = \frac{-\Delta[\text{Br}_2]}{\Delta t} = - \frac{[\text{Br}_2]_f - [\text{Br}_2]_i}{t_f - t_i}$$

$$\text{average rate} = - \frac{0.00500 - 0.0101}{250.0 - 50.0} = 2.55 \times 10^{-5} \text{ M.s}^{-1}$$





Time (s)	[Br ₂] (M)
0.0	0.0120
50.0	0.0101
100.0	0.00846
150.0	0.00710
200.0	0.00596
250.0	0.00500
300.0	0.00420
350.0	0.00353
400.0	0.00296



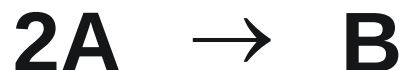
instantaneous rate: rate at specific instance in time

instantaneous rate at 200 s = $2.09 \times 10^{-5} \text{ M/s}$

initial rate: rate at $t = 0 \text{ s} = 0.0120 \text{ M/s}$



Reaction Rates & Stoichiometry



$$\text{Rate} = \frac{\Delta[B]}{\Delta t}$$

rate of disappearance of A =

2 x rate of appearance of B

$$\text{rate} = - \frac{\Delta[A]}{\Delta t} = 2 \times \frac{\Delta[B]}{\Delta t}$$

$$\text{rate} = - \frac{1}{2} \frac{\Delta[A]}{\Delta t} = \frac{\Delta[B]}{\Delta t}$$



Reaction Rates & Stoichiometry

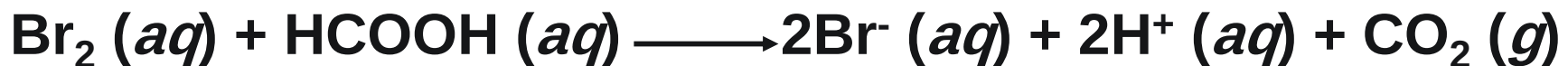


$$\begin{aligned}\text{rate} &= -\frac{1}{a} \frac{\Delta[A]}{\Delta t} = -\frac{1}{b} \frac{\Delta[B]}{\Delta t} \\ &= \frac{1}{c} \frac{\Delta[C]}{\Delta t} = \frac{1}{d} \frac{\Delta[D]}{\Delta t}\end{aligned}$$



$$\text{rate} = -\frac{1}{2} \frac{\Delta[\text{N}_2\text{O}_5]}{\Delta t} = \frac{1}{4} \frac{\Delta[\text{NO}_2]}{\Delta t} = \frac{\Delta[\text{O}_2]}{\Delta t}$$





Time (s)	[Br ₂] (M)	
0.0	0.0120	
50.0	0.0101	
100.0	0.00846	2.96×10^{-5}
150.0	0.00710	1.42
200.0	0.00596	2.09×10^{-5}
250.0	0.00500	1.42
300.0	0.00420	1.48×10^{-5}
350.0	0.00353	
400.0	0.00296	

rate $\propto [\text{Br}_2]^1$

2

2

14.3 Concentration and Rate Law

How the rate of the reaction changes with concentration of REACTANTS



$$\text{rate} \propto [A]^x$$

$$\text{rate} \propto [B]^y$$

$$\text{rate} = k [A]^x [B]^y$$

RATE LAW

k = rate constant

depends on: Specific reaction

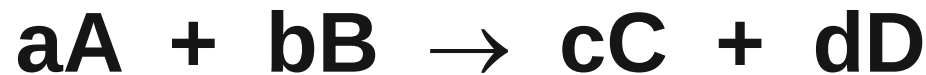
Temperature

Presence of catalyst

larger k : faster reaction

Units: depend on x & y





$$\text{rate} = k [A]^x [B]^y$$

RATE LAW

x and y:

x is the order of the reaction with respect to A

y is the order of the reaction with respect to B

x + y is the overall order of the reaction

Not related to stoichiometric coefficients in the balanced equation

Determined only experimentally

Mostly positive small integers (0, 1, 2)

May be negative or fractions





$$\text{rate} = k[A]^x \quad \text{RATE LAW}$$

Reaction order

Unit of k

If $x = 1$

first

s^{-1}

If $x = 2$

2nd

$s^{-1} M^{-1}$

If $x = 3$

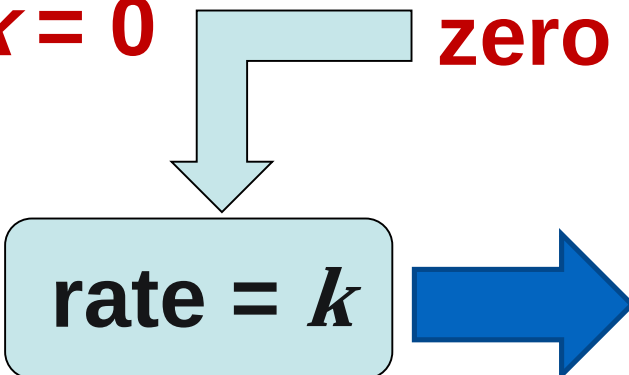
3nd

$s^{-1} M^{-2}$

If $x = 0$

zero

$s^{-1} M$


$$\text{rate} = k$$

Rate of reaction is initial
reactant concentration
independent



Rate Law From Initial Rates



$$\text{rate} = k [\text{F}_2]^x [\text{ClO}_2]^y$$

Rate Data for the Reaction Between F_2 and ClO_2

	$[\text{F}_2] (M)$	$[\text{ClO}_2] (M)$	Initial Rate (M/s)
1	0.10	0.010	1.2×10^{-3}
2	0.10	0.040	4.8×10^{-3}
3	0.20	0.010	2.4×10^{-3}

$\frac{0.010}{0.040} = 0.25$ $\frac{1.2 \times 10^{-3}}{4.8 \times 10^{-3}} = 0.25$

$y = 1$

1st order in ClO_2





$$\text{rate} = k [\text{F}_2]^x [\text{ClO}_2]^y$$

	$[\text{F}_2] (M)$	$[\text{ClO}_2] (M)$	Initial Rate (M/s)
1	0.10	0.010	1.2×10^{-3}
2	0.10	0.040	4.8×10^{-3}
3	0.20	0.010	2.4×10^{-3}

$= 0.5$ (for rows 1 and 2)
 $= 0.5$ (for rows 1 and 3)

$x = 1$

1st order in F_2

$$\text{rate} = k [\text{F}_2]^x [\text{ClO}_2]^y$$

	$[\text{F}_2] (M)$	$[\text{ClO}_2] (M)$	Initial Rate (M/s)
1	0.10	0.010	1.2×10^{-3}
2	0.10	0.040	4.8×10^{-3}
3	0.20	0.010	2.4×10^{-3}

$$\frac{\text{rate}_2}{\text{rate}_1} = \frac{\cancel{k} [\text{F}_2]_2^x [\text{ClO}_2]_2^y}{\cancel{k} [\text{F}_2]_1^x [\text{ClO}_2]_1^y} = \frac{4.8 \times 10^{-3}}{1.2 \times 10^{-3}}$$

$$4 = \frac{\cancel{(0.10)}^x (0.040)^y}{\cancel{(0.10)}^x (0.010)^y} = \frac{(0.040)^y}{(0.010)^y} = (4)^y$$

$$y = 1$$

1st order in ClO_2





	$[\text{F}_2] (M)$	$[\text{ClO}_2] (M)$	Initial Rate (M/s)
1	0.10	0.010	1.2×10^{-3}
2	0.10	0.040	4.8×10^{-3}
3	0.20	0.010	2.4×10^{-3}

$$\frac{\text{rate}_3}{\text{rate}_1} = \frac{k [\text{F}_2]_3^x [\text{ClO}_2]_3^y}{k [\text{F}_2]_1^x [\text{ClO}_2]_1^y} = \frac{2.4 \times 10^{-3}}{1.2 \times 10^{-3}}$$

$$2 = \frac{(0.20)^x (0.010)^y}{(0.10)^x (0.010)^y} = \frac{(0.20)^x}{(0.10)^x} = (2)^x$$

$$x = 1 \quad 1^{\text{st}} \text{ order in } \text{F}_2$$

$$\text{rate} = k[\text{F}_2]^1[\text{ClO}_2]^1$$

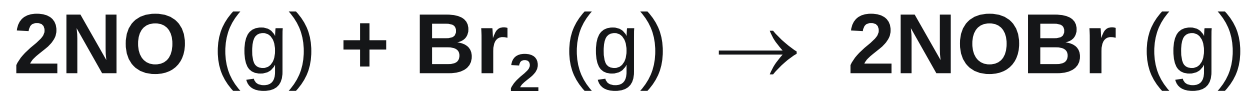
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Rate when:

Rate Law	$x = 1$	$x = 2$	$x = 3$	$x = 0$
$k[A]^x$	$k[A]$	$k[A]^2$	$k[A]^3$	$k[A]^0 = k$
$k[2A]^x$	$2k[A]$	$4k[A]^2$	$8k[A]^3$	$k[2A]^0 = k$
$k[3A]^x$	$3k[A]$	$9k[A]^2$	$27k[A]^3$	$k[3A]^0 = k$
Order in A	1 st	2 nd	3 rd	0 th





	[NO] (M)	[Br₂] (M)	Initial Rate (M/s)
1	0.10	0.10	12
2	0.10	0.20	24
3	0.30	0.10	108

Determine the rate law

$$\text{rate} = k[\text{NO}]^x[\text{Br}_2]^y$$

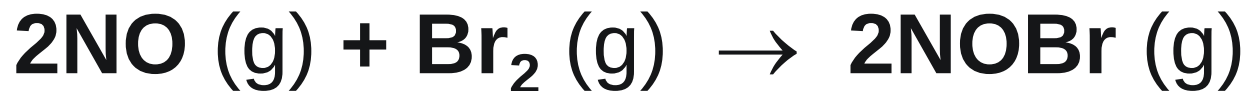
$$\frac{\text{rate}_2}{\text{rate}_1} = \frac{24}{12} = \frac{\cancel{k} \cancel{(0.10)}^x (0.20)^y}{\cancel{k} \cancel{(0.10)}^x (0.10)^y} = (2)^y$$

$$2 = (2)^y$$

$$y = 1$$

1st order in Br₂





	[NO] (M)	[Br₂] (M)	Initial Rate (M/s)
1	0.10	0.10	12
2	0.10	0.20	24
3	0.30	0.10	108

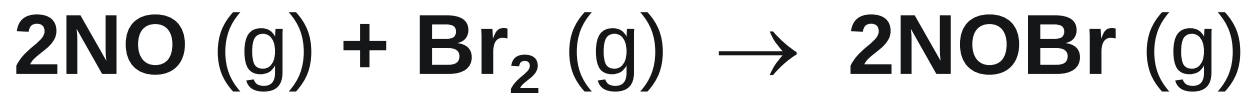
Determine the rate law **rate = $k[\text{NO}]^x[\text{Br}_2]^y$**

$$\frac{\text{rate}_3}{\text{rate}_1} = \frac{108}{12} = \frac{\cancel{k} (0.30)^x \cancel{(0.10)^y}}{\cancel{k} (0.10)^x \cancel{(0.10)^y}} = (3)^x$$

$$9 = (3)^x \quad x = 2 \quad \text{2nd order in NO}$$

$$\text{rate} = k[\text{NO}]^2 [\text{Br}_2]$$





	[NO] (M)	[Br₂] (M)	Initial Rate (M/s)
1	0.10	0.10	12
2	0.10	0.20	24
3	0.30	0.10	108

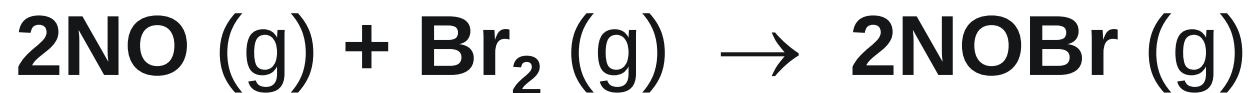
Calculate the rate constant for the reaction

$$\text{rate} = k [\text{NO}]^2 [\text{Br}_2]$$

$$k = \frac{\text{rate}}{[\text{NO}]^2 [\text{Br}_2]} = \frac{108 \cancel{\text{M}} \cdot \text{s}^{-1}}{(0.30 \text{ M})^2 (0.10 \cancel{\text{M}})}$$

$$k = 1.2 \times 10^4 \text{ M}^2 \cdot \text{s}^{-1}$$





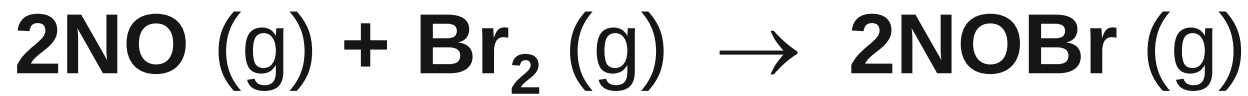
What is the rate of the reaction when $[\text{NO}] = 0.15 \text{ M}$ and $[\text{Br}_2] = 0.25 \text{ M}$

$$\text{rate} = k [\text{NO}]^2 [\text{Br}_2]$$

$$\text{rate} = (1.2 \times 10^4 \text{ M}^2.\text{s}^{-1})(0.15 \text{ M})^2(0.25 \text{ M})$$

$$\text{rate} = 67.5 \text{ M}.\text{s}^{-1}$$





How is the appearance of NOBr related to the rate of disappearance Br₂ and NO

$$\frac{1}{2} \frac{\Delta[\text{NOBr}]}{\Delta t} = - \frac{1}{2} \frac{\Delta[\text{NO}]}{\Delta t} = - \frac{\Delta[\text{Br}_2]}{\Delta t}$$

therefore

$$\frac{\Delta[\text{NOBr}]}{\Delta t} = - \frac{\Delta[\text{NO}]}{\Delta t} = - \frac{2 \Delta[\text{Br}_2]}{\Delta t}$$



Change of Concentration with Time

Rate law:

Tells how the reaction rate depends on reactant concentration

How the concentrations of reactants and products change with time?



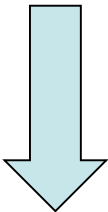
Change of Concentration with Time

The Integrated Rate Law:

1. First-Order Reactions



$$\text{rate} = k[A]^1 \longleftrightarrow \text{rate} = - \frac{\Delta[A]}{\Delta t}$$

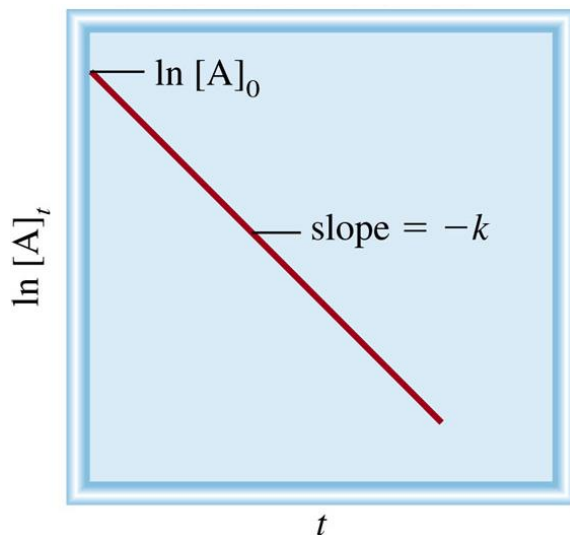

$$- \frac{\Delta[A]}{\Delta t} = k[A]$$



$$\frac{d[A]}{[A]} = -k dt \quad \longrightarrow \quad \int_{[A]_0}^{[A]_t} \frac{d[A]}{[A]} = -k \int_0^t dt$$

linear

$$\ln \frac{[A]_t}{[A]_0} = -kt$$



$$\ln[A]_t = -kt + \ln[A]_0$$



y



= - a x



+ b



slope



intercept

For gas-phase reactions:

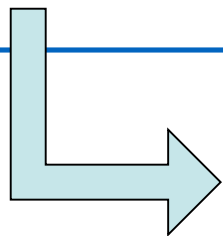
$$\ln P_t = -k t + \ln P_0$$

Reaction Half-Life $t_{1/2}$

The time required for the concentration of a reactant to decrease to half of its initial concentration

$$t = \frac{\ln \frac{[A]_0}{[A]_t}}{k} \quad \text{at } t_{1/2}: [A]_t = \frac{[A]_0}{2}$$

$$t_{1/2} = \frac{\ln 2}{k} = \frac{0.693}{k}$$



**constant
independent of $[A]_0$**

2. Second-Order Reactions



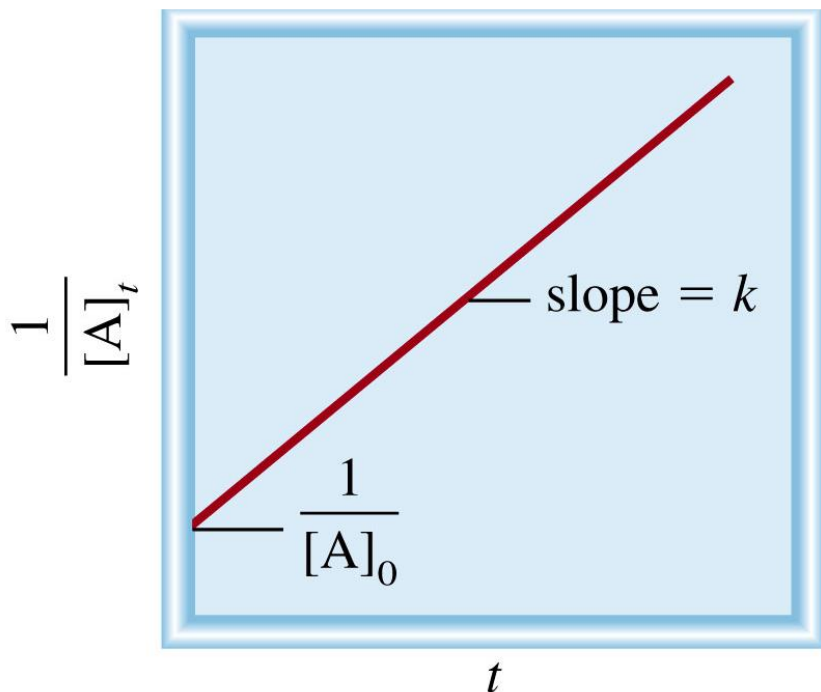
$$\text{rate} = k[A]^2 \longleftrightarrow \text{rate} = - \frac{\Delta[A]}{\Delta t}$$

$$- \frac{\Delta[A]}{\Delta t} = k[A]^2$$

$$\frac{d[A]}{[A]^2} = - k dt \longrightarrow \int_{[A]_0}^{[A]_t} \frac{d[A]}{[A]^2} = - k \int_0^t dt$$



linear



$$\frac{\underset{\substack{\text{red } y \\ \downarrow}}{1}}{\underset{\substack{\text{blue } b \\ \downarrow}}{[A]_t}} = \frac{1}{\underset{\substack{\text{blue } b \\ \downarrow}}{[A]_0}} + \underset{\substack{\text{blue } m \\ \downarrow}}{k} \underset{\substack{\text{red } x \\ \downarrow}}{t}$$

For gas-phase reactions:

$$\frac{1}{P_t} = k t + \frac{1}{P_0}$$

Reaction Half-Life Time $t_{1/2}$

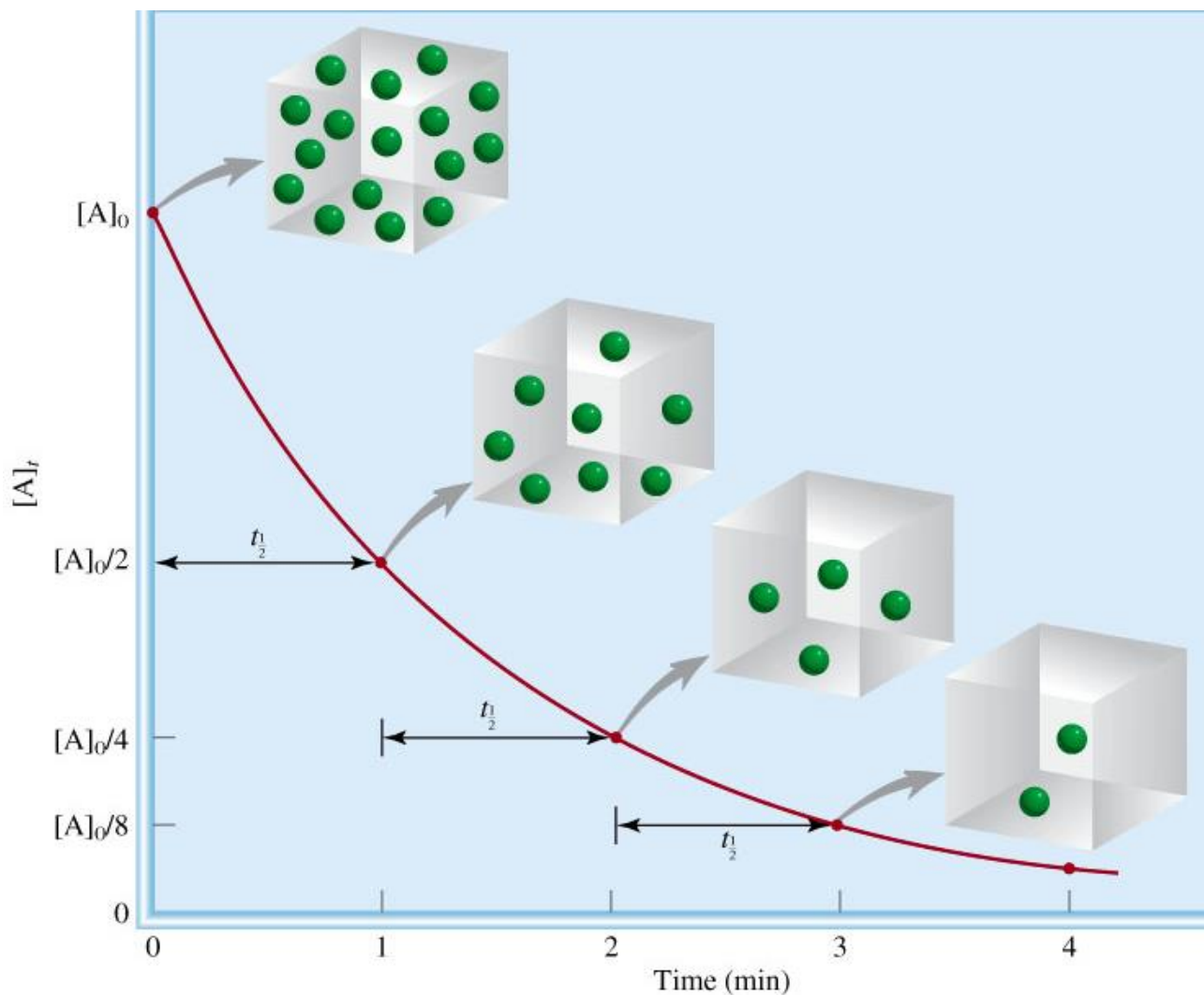
$$\frac{1}{[A]_t} = k t + \frac{1}{[A]_0} \longrightarrow \frac{1}{[A]_0/2} = k t_{1/2} + \frac{1}{[A]_0}$$

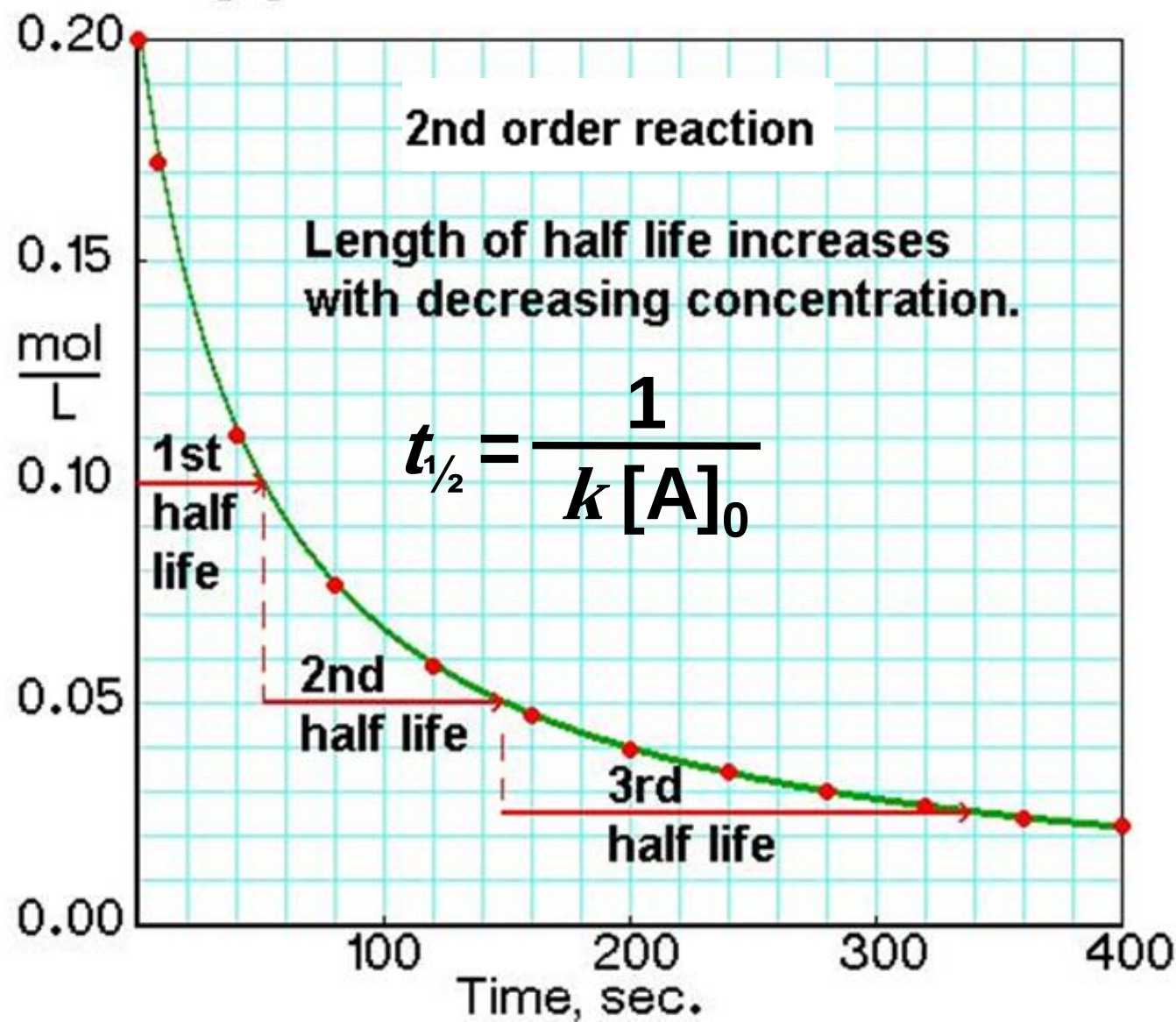
Depends on initial
concentration and k

$$t_{1/2} = \frac{1}{k [A]_0}$$



half life for a first order reaction is constant





The rate constant for the reaction $2A \rightarrow B$ is $2.8 \times 10^{-2} \text{ s}^{-1}$ at 80°C . A) **How long** will it take for A to decrease from 0.88 M to 0.14 M ?

$$\ln[A]_t = -k t + \ln[A]_0$$

$[A]_0$	$\leftarrow$	$2A \rightarrow$	B
at $t = 0$		0.88	0.0
at $t = ??$		0.14	$\rightarrow [A]_t$

$$t = \frac{\ln[A]_0 - \ln[A]_t}{k}$$

$$t = \frac{\ln \frac{[A]_0}{[A]_t}}{k} = \frac{\ln \frac{0.88 \text{ M}}{0.14 \text{ M}}}{2.8 \times 10^{-2} \text{ s}^{-1}} = 66 \text{ s}$$



The rate constant for the reaction $2A \rightarrow B$ is $2.8 \times 10^{-2} \text{ s}^{-1}$ at 80°C .

B) What is the **half-life** for the reaction?

Answer: 24.6 s

C) **How long** will it take for A to decrease from 0.88 M to 0.11 M ?

Answer: 73.9 s

D) **How long** will it take convert 75% of A to product?

Answer: 49.2 s



The rate constant for the reaction $2A \rightarrow B$ is $2.8 \times 10^{-2} \text{ s}^{-1} \text{ M}^{-1}$ at 80°C .

A) How long will it take for A to decrease from 0.88 M to 0.14 M ?

Answer: 214.5 s

B) What is the half-life for the reaction?

Answer: 40.6 s

C) How long will it take for A to decrease from 0.88 M to 0.11 M ?

Answer: 284.2 s

D) How long will it take convert 75% of A to product?

Answer: 122 s



3. Zero-Order Reactions



$$\text{rate} = k[A]^0 \longleftrightarrow - \frac{\Delta[A]}{\Delta t} = \text{rate}$$

$$k = - \frac{\Delta[A]}{\Delta t} \longrightarrow d[A] = - k dt$$

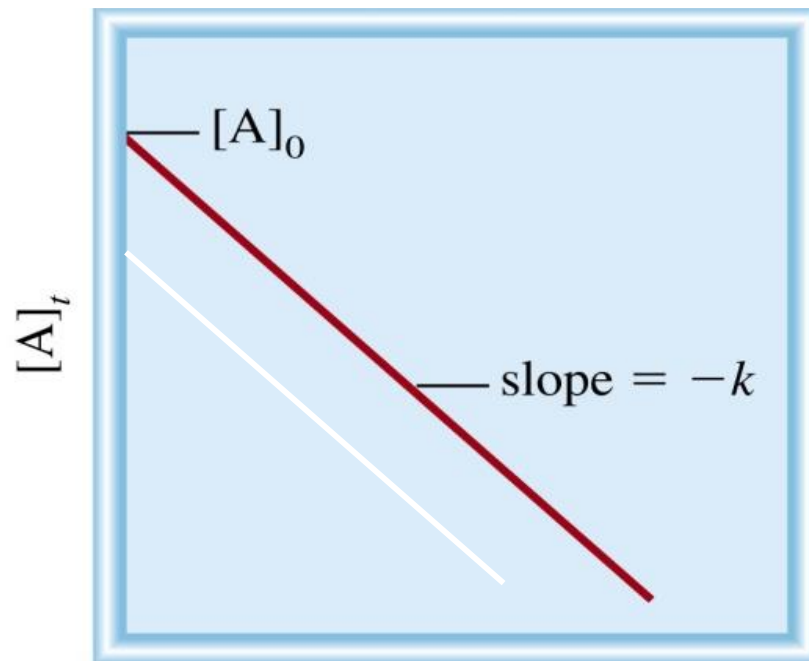
$$\int_{[A]_0}^{[A]_t} d[A] = - k \int_0^t dt$$

$$[A]_t = - k t + [A]_0$$

y

$m x$

b



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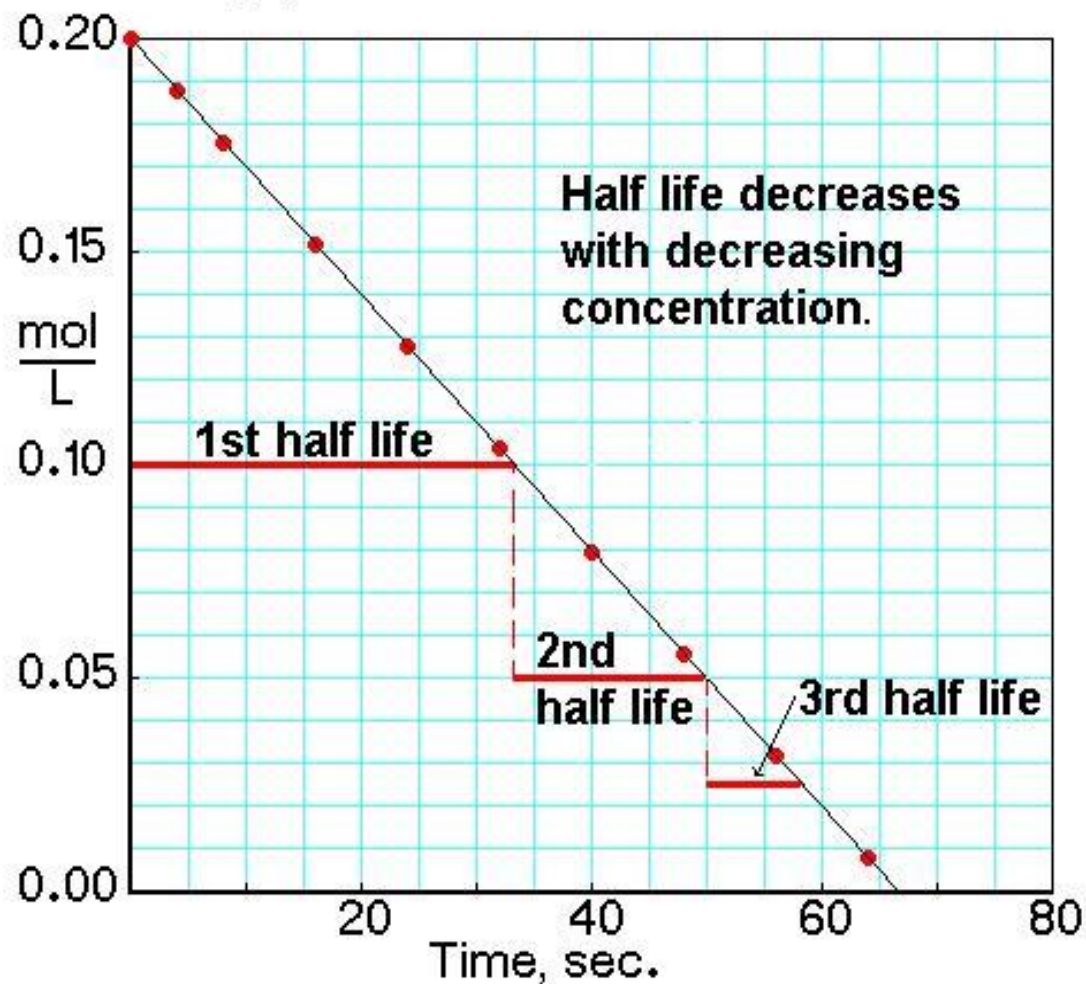
Reaction Half-Life Time

$$[A]_t = -k t + [A]_0$$

$$\frac{[A]_0}{2} = -k t_{1/2} + [A]_0$$

$$t_{1/2} = \frac{[A]_0}{2k}$$

[A] vs time for a 0 order reaction



Order	Rate Law	$\int$ Rate Law	Slope	Intercept	$t_{1/2}$
0 th	Rate = k	$[A]_t = -kt + [A]_0$	- k M.s ⁻¹	$[A]_0$	$\frac{[A]_0}{2k}$
1 st	Rate = $k[A]$	$\ln[A]_t = -kt + \ln[A]_0$	- k s ⁻¹	$\ln[A]_0$	$\frac{0.693}{k}$
2 nd	Rate = $k[A]^2$	$\frac{1}{[A]_t} = kt + \frac{1}{[A]_0}$	k M ⁻¹ .s ⁻¹	$\frac{1}{[A]_0}$	$\frac{1}{k[A]_0}$



The following initial rate data were obtained for the stoichiometric reaction: $3A + B \rightarrow 2P$

Exp.	$[A]_0, M$	$[B]_0, M$	Initial rate = $-d[A]/dt$
1	0.20	0.20	1.2×10^{-8}
2	0.20	0.60	1.2×10^{-8}

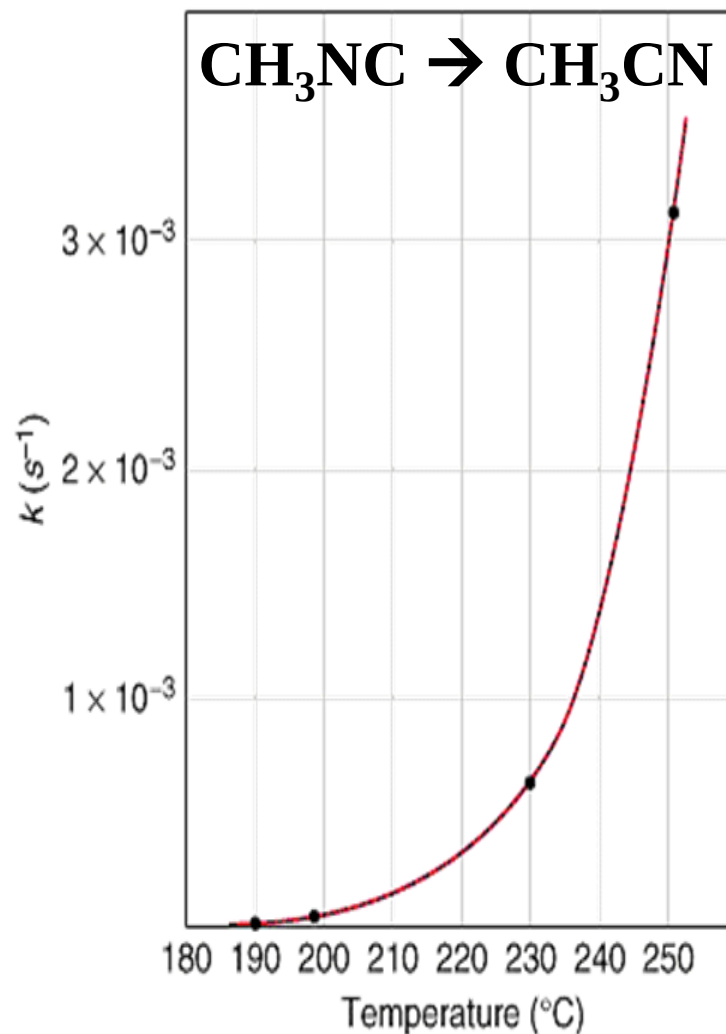
For a third experiment, a plot of $1/[A]$ versus time was found to be linear. What is the order of the reaction with respect to the concentration of A and B?

	Order with Respect to $[A]$	Order with Respect to $[B]$
A)	1	0 
B)	2	1
C)	2	0 
D)	0	2



14.5 Temperature and Rate

- The rates of most chemical reactions increases as temperature rises.
- The faster rate at high temperature is due to an increase in the rate constant with increasing temperature.
- All explained by the collision model theory.



The Collision Model

➤ Molecules must collide to react.

Rate

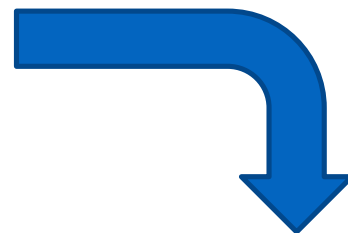
$\propto$

**Number of collision
per second**

As reactant
conc. ↑



Number of
collisions ↑



As Temp. ↑



Molecular
speeds ↑

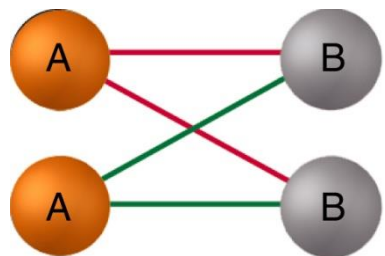


Molecules collide
more forcefully and
more frequently



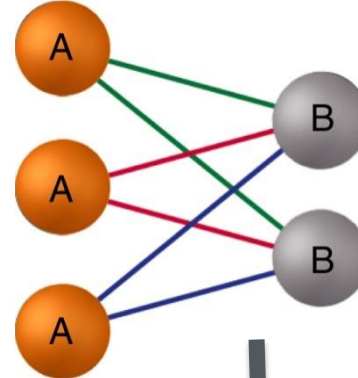
**Reaction
Rate ↑**





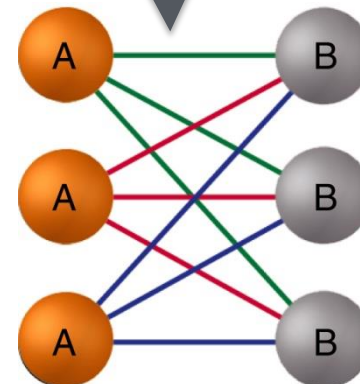
4 collisions

add another

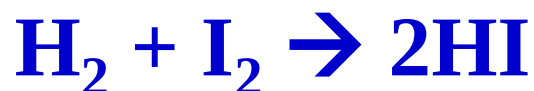


6 collisions

add another



9 collisions



each molecule undergoes
about 10^{10} collisions/second

The reaction expected to be over
in less than a second

Actually, the reaction proceeds very
slowly, **Why?**

Only small fraction of collisions produce reaction. Why?

1. Colliding molecules should have a total kinetic energy equal to or greater than a minimum value (threshold) called the **activation energy, E_a** .
2. The relative **orientation** of the reactants must allow the formation of any new bonds necessary to produce products (steric effect)



1. The Orientation Factor

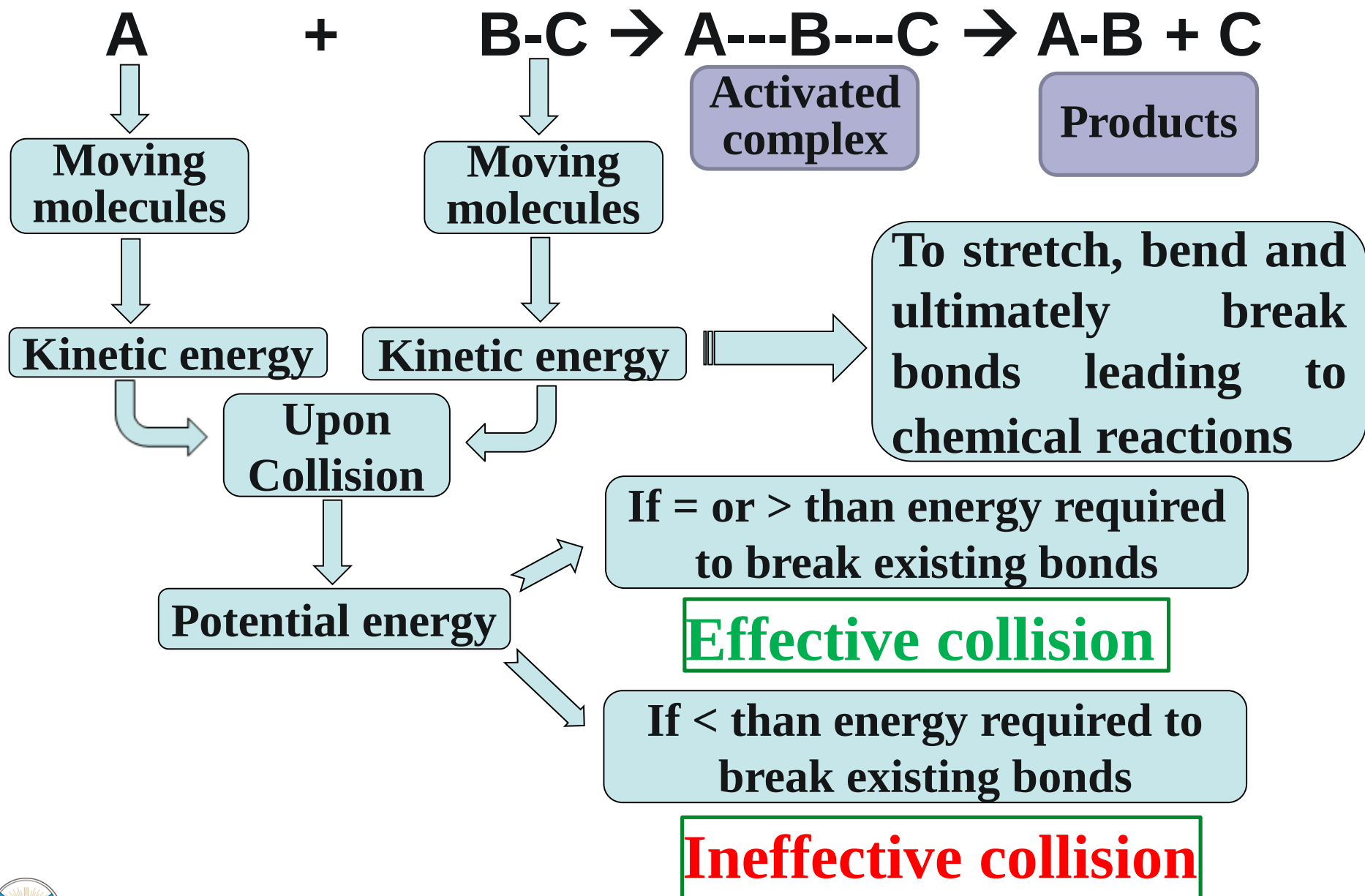


Effective collision $\longrightarrow$ **Reaction occurs**



Ineffective collision $\longrightarrow$ **No reaction**

2. Activation Energy (E_a)



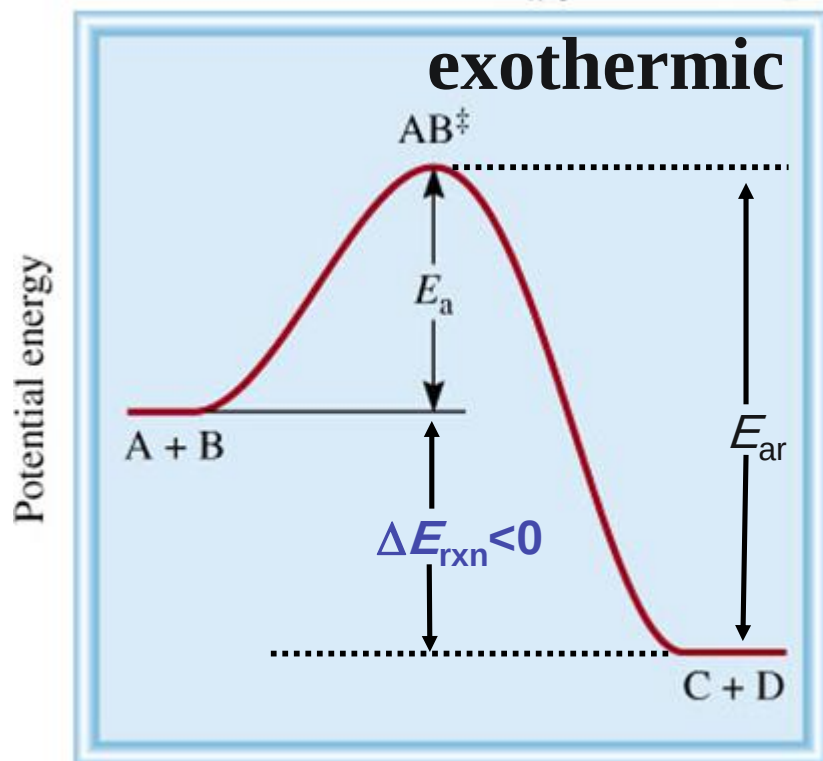
2. Activation Energy (E_a)

- ❖ **Activation energy, E_a** , is the minimum energy required to initiate a chemical reaction and its value varies from reaction to reaction.
- ❖ The rate depends on the magnitude of E_a , generally the lower the value of E_a is, the faster the reaction.

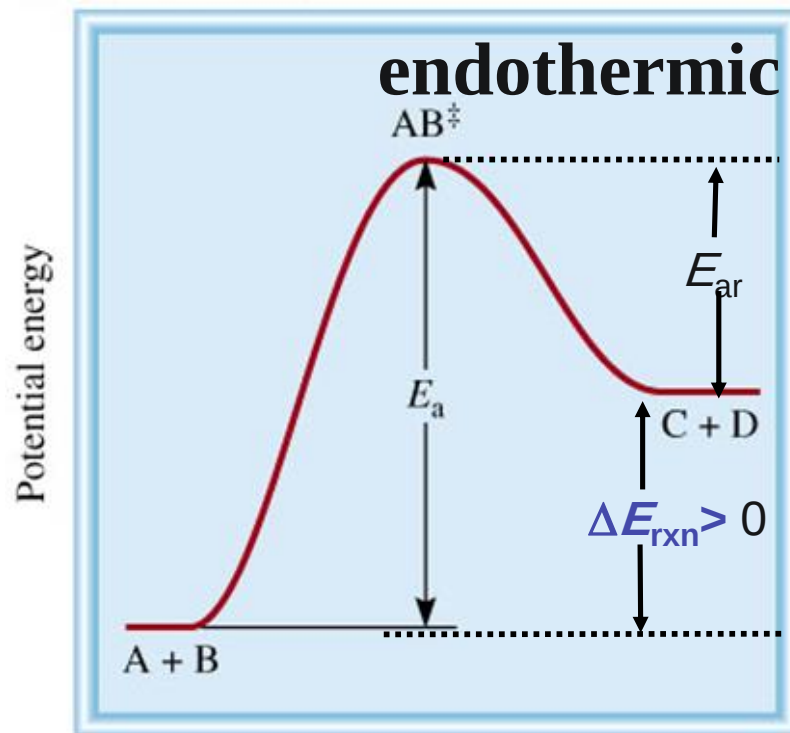




$AB^\ddagger$: activated complex or transition state



Reaction progress



Reaction progress

$$E_a = P.E_{AB^\ddagger} - P.E_{A+B}$$

$$E_{ar} = P.E_{AB^\ddagger} - P.E_{C+D}$$

$$\Delta E_{rxn} = P.E_{C+D} - P.E_{A+B}$$

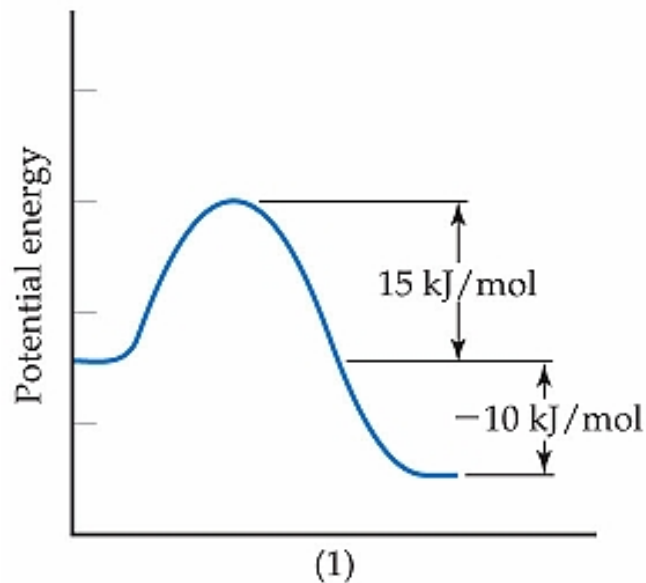
or

$$\Delta E_{rxn} = E_a - E_{ar}$$

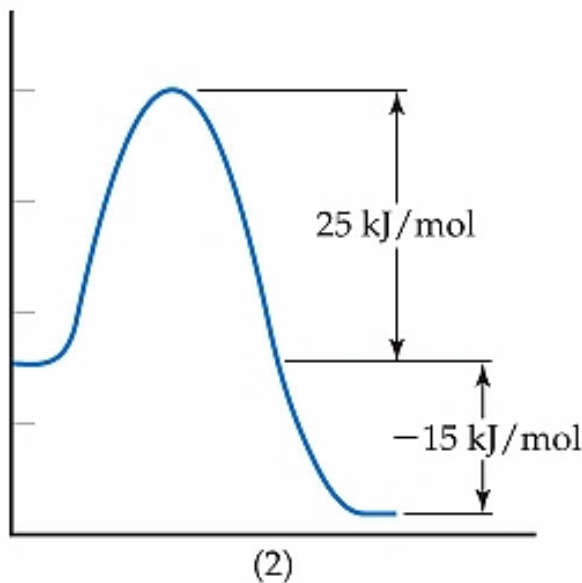
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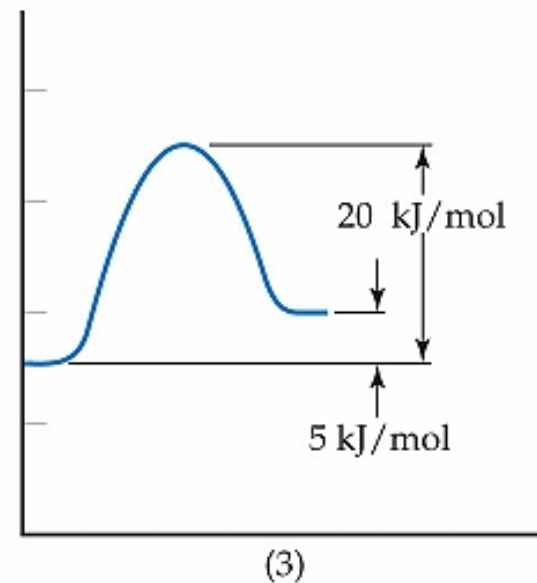
Consider a series of reactions having these energy profiles. Rank the reactions from slowest to fastest assuming that they have nearly the same value for the frequency factor A .



$$E_a = 15 \text{ kJ/mol}$$



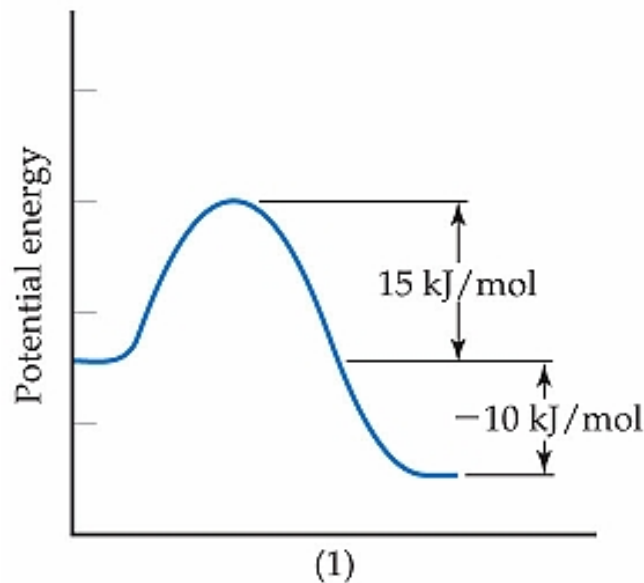
$$25 \text{ kJ/mol}$$



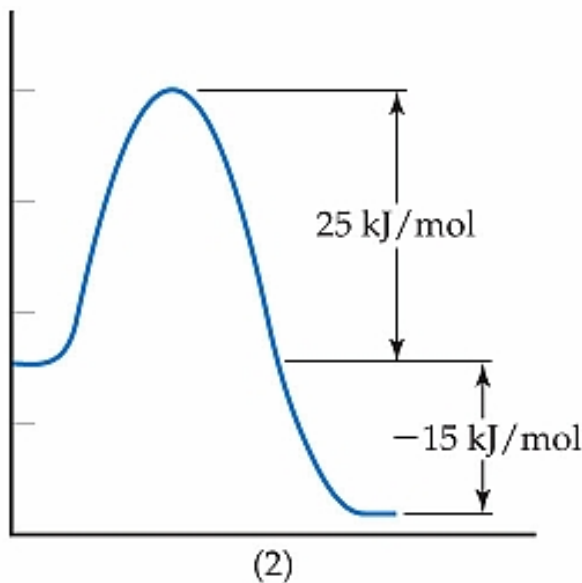
$$20 \text{ kJ/mol}$$

$$2 < 3 < 1$$

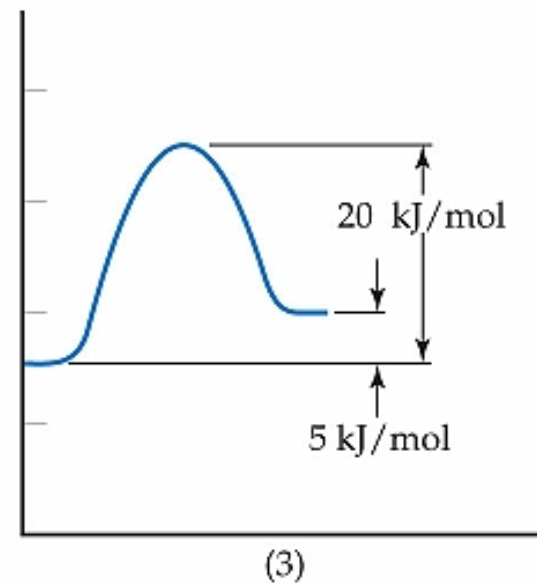
Consider a series of reactions having these energy profiles. Rank the reverse reactions from slowest to fastest assuming that they have nearly the same value for the frequency factor A .



$$E_a = 25 \text{ kJ/mol}$$



$$40 \text{ kJ/mol}$$



$$15 \text{ kJ/mol}$$

$$2 < 1 < 3$$

Temperature and Collision Energy

- An increase in temperature causes an increase in the kinetic energy of the molecules.
- This leads to an increase in the *fraction* of the molecules have kinetic energy equal to or greater than E_a , which enhances the rate of reaction.
- The *fraction* is given by the expression

$$f = e^{-E_a/RT}$$

R = the gas constant (8.314 J/mol.K)

T = absolute temperature (K)



Arrhenius Equation

Rate $\propto$ Fraction of effective collision

$$\text{Rate} \propto e^{-E_a / RT}$$

$$k = Ae^{-E_a / RT}$$

k = rate constant

A = constant; frequency factor (frequency of properly oriented collisions)

$$\text{rate} \propto T$$

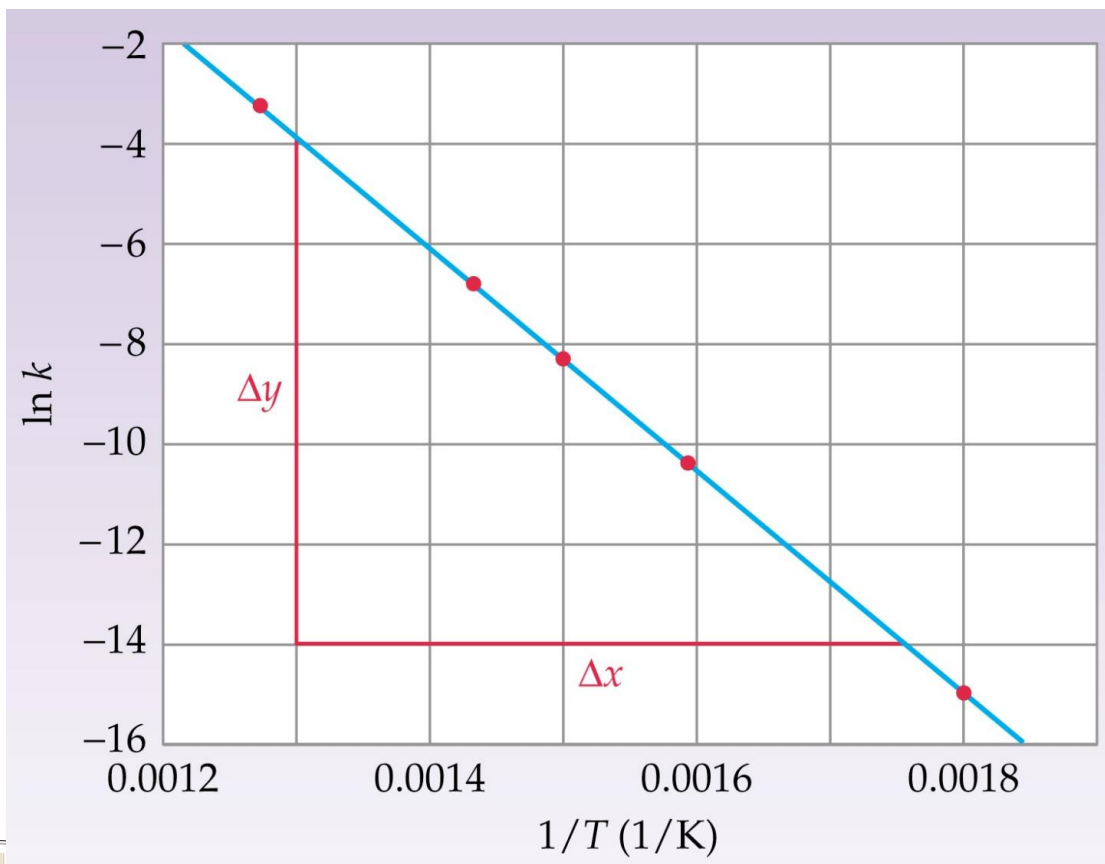
$$\text{rate} \propto \frac{1}{E_a}$$



Determining the Activation Energy

$$k = Ae^{-E_a / RT} \quad \longrightarrow \quad \ln k = - \frac{E_a}{R} \frac{1}{T} + \ln A$$

$\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$
y **m** **x** **b**



$$\text{slope} = -\frac{E_a}{R}$$

$$\text{intercept} = \ln A$$

Arrhenius Equation

At two different temperatures:

$$\ln k_1 = -\frac{E_a}{RT_1} + \ln A$$

$$\ln k_2 = -\frac{E_a}{RT_2} + \ln A$$

Subtract & rearrange

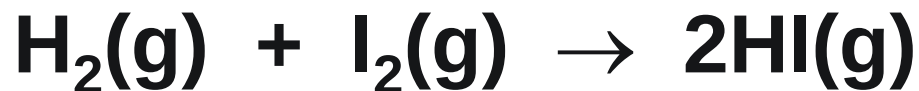
$$\ln \frac{k_1}{k_2} = \frac{E_a}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$



Ex: A certain first order reaction has a rate constant of $2.75 \times 10^{-2} \text{ s}^{-1}$ at 20°C . What is the **value of “k” at 60°C** if $E_a = 75.5 \text{ kJ/mole}$.

Answer = 1.1 s^{-1}





The rate law for this reaction is first order in each of the two reactants and has a $k = 0.0028/\text{M.s}$ at 200°C measured for $-\text{d}[\text{H}_2]/\text{dt}$. E_a for the reaction is $170. \text{ kJ/mol}$. What is the rate of HI formation at 300°C when both H_2 and I_2 are at a concentration of 0.0150 M ?

$$\text{rate} = \frac{-\Delta[\text{H}_2]}{\Delta t} = k[\text{H}_2][\text{I}_2]$$

At 300°C (573 K):

$$\frac{\Delta[\text{HI}]}{\Delta t} = 2 \frac{-\Delta[\text{H}_2]}{\Delta t} = 2 k_{573} [\text{H}_2][\text{I}_2]$$

55



$$\ln \frac{k_1}{k_2} = \frac{E_a}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

$$\ln \frac{k_{473}}{k_{573}} = \frac{E_a}{R} \left(\frac{1}{573} - \frac{1}{473} \right)$$

$$\ln \frac{0.0028 \text{ M}^1.\text{s}^{-1}}{k_{573}} = \frac{1.70 \times 10^5 \text{ J/mol}}{8.314 \text{ J.mol}^{-1}.\text{K}^{-1}} (-3.69 \times 10^{-4} \text{ K}^{-1})$$

$$k_{573} = 5.29 \text{ M}^1.\text{s}^{-1}$$

$$\begin{aligned} \frac{\Delta[\text{HI}]}{\Delta t} &= 2k_{573}[\text{H}_2][\text{I}_2] = (2) (5.29) (0.0150)^2 \\ &= 0.00238 \text{ M.s}^{-1} \end{aligned}$$



14.5 Reaction Mechanisms

- The steps by which a reaction occurs.
- Reactions may occur all at once (in a single step) known as **elementary reaction**.
- Or occurs by a multistep mechanism consisting of a sequence of elementary reactions.



Two molecules participate as reactants



Bimolecular

2nd order rxn

- **Molecularity:** the number of reactant molecules involved in an elementary reaction.
- *If a reaction is elementary, its rate law is based on its molecularity.*

Molecularity	Elementary rxn	Rate Law	Rxn. order
unimolecular	$A \rightarrow P$	$\text{rate} = k[A]$	1 st
bimolecular	$A + A \rightarrow P$	$\text{rate} = k[A]^2$	2 nd
	$A + B \rightarrow P$	$\text{rate} = k[A][B]$	
termolecular	$A + A + A \rightarrow P$	$\text{rate} = k[A]^3$	3 rd
	$A + A + B \rightarrow P$	$\text{rate} = k[A]^2[B]$	
	$A + B + C \rightarrow P$	$\text{rate} = k[A][B][C]$	

- **Unimolecular:** an elementary step that involves one reactant molecule.
- **Bimolecular:** an elementary step that involves two reactant molecules (may be of the same or different types).
- **Termolecular:** an elementary step that involves three reactant molecules (may be of the same or different types). (fairly rare).

The rxn. order for each reactant in an elementary rxn. is equal to its stoichiometric coefficient in the chemical equation for that step.

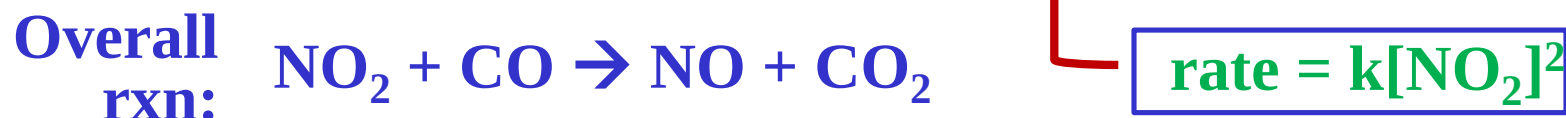
Sample Exercise



order $\neq$ stiochiometric coefficient $\rightarrow$ multistep rxn

Rate law

Molecularity



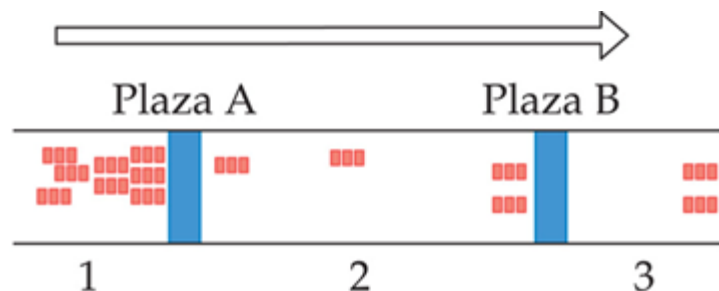
- Steps of a mechanism must satisfy two requirements
 - Sum of elementary steps must equal the overall balanced equation.
 - The rate law must have same rate law as determined from experimental data.
- NO_3 is an intermediate in the reaction mechanism
 - Intermediate:** a substance that is produced in an early step and consumed in a later step.

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Rate Determining Step

- The rate law for multistep rxn is given by Rate Determining Step (RDS).

RDS is the slowest step in the mechanism

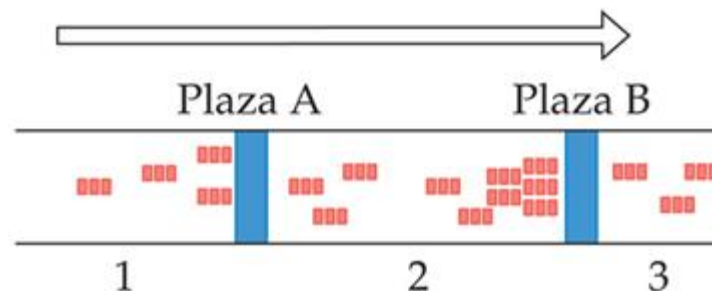


Cars slowed at plaza A

Step 1: slow

Step 2: fast

Step 1: RDS



Cars slowed at plaza B

Step 1: fast

Step 2: slow

Step 2: RDS

Step 1: Point 1 → Point 2

(Through plaza A)

Step 2: Point 2 → Point 3

(Through plaza B)

- Point 2 is an intermediate point.

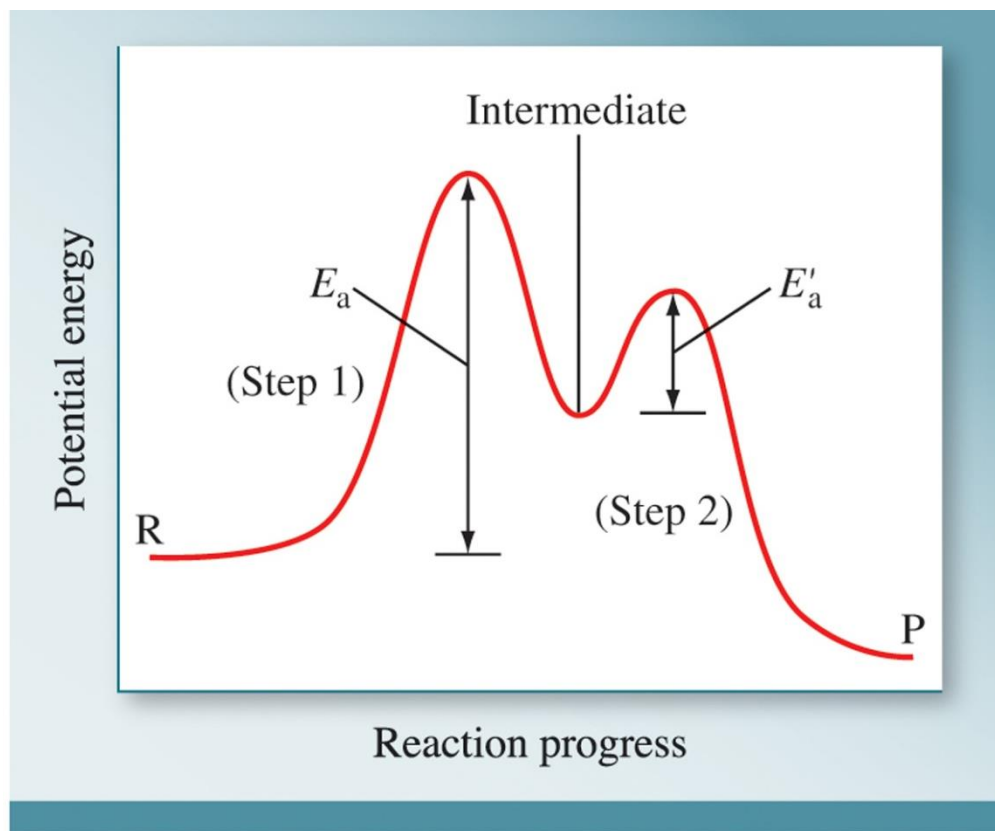
Sample Exercise

The decomposition of hydrogen peroxide may occur in the following two steps



- If step 1 is the rate-determining step, then the rate law is
rate = $k_1[\text{H}_2\text{O}_2][\text{I}^-]$
- IO^- is an intermediate.
- I^- is present at the start of the reaction and is present at the end of the reaction $\Rightarrow$ $\boxed{\text{I}^- \text{ serves as a catalyst}}$
- I^- does not appear in the overall balanced equation.

Potential Energy Diagram



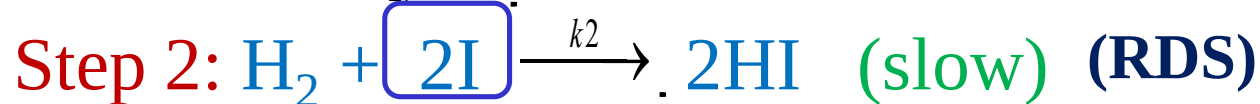
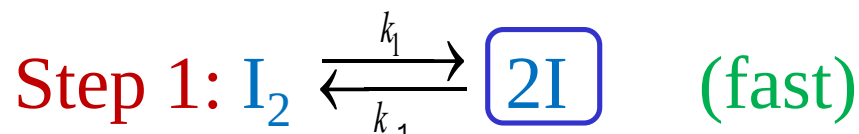
- $E_{a1} < E_{a2} \longrightarrow$ 2nd step faster than the 1st step
Exothermic rxn.

Sample Exercise

Given overall equation: $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightarrow 2\text{HI}(\text{g})$

Experimentally the rate law is: $\text{rate} = k[\text{H}_2][\text{I}_2]$

Two step mechanism was proposed:



- I is an intermediate and should not appear in the rate expression
 $\text{rate} = k_2[\text{H}_2][\text{I}]^2$

- Consider the first equilibrium step:
the forward rate = the reverse rate

$$k_1[\text{I}_2] = k_{-1}[\text{I}]^2$$

$$k_1/k_{-1}[\text{I}_2] = [\text{I}]^2$$

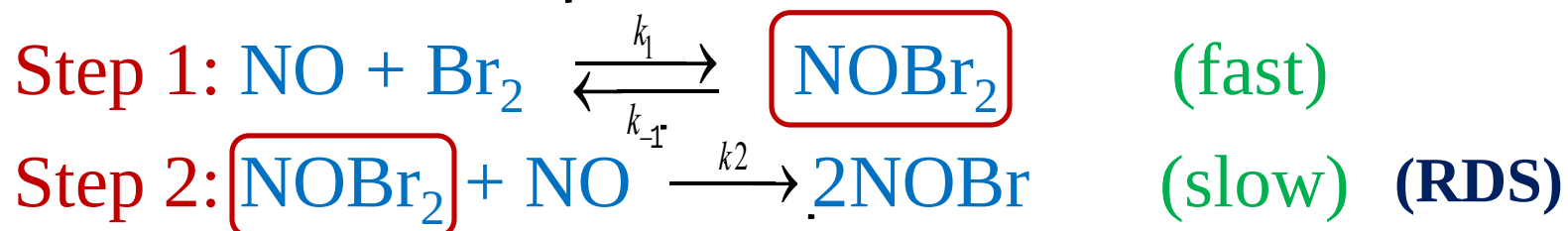
$$\text{rate} = \{k_2 k_1/k_{-1}\} [\text{I}_2][\text{H}_2]$$

$$k = k_2 k_1/k_{-1}$$

Sample Exercise

The rate law for the reaction $2\text{NO} + \text{Br}_2 \rightarrow 2\text{NOBr}$ is found (experimentally) to be $\text{rate} = k[\text{NO}]^2[\text{Br}_2]$

Show that the following proposed mechanism produces a rate law consistent with the experimentally proposed one.



- The rate law for the rxn = the rate law for the slowest step

$$\text{rate} = k_2[\text{NO}][\text{NOBr}_2]$$

- NOBr_2 is an intermediate and should not appear in the rate expression

- Consider the first equilibrium step:
the forward rate = the reverse rate

$$k_1 [\text{NO}] [\text{Br}_2] = k_{-1} [\text{NOBr}_2]$$

- Solving for $[\text{NOBr}_2]$ gives us

$$\frac{k_1}{k_{-1}} [\text{NO}] [\text{Br}_2] = [\text{NOBr}_2]$$

- Substituting this expression for $[\text{NOBr}_2]$ in the rate law for the rate-determining step gives

$$\text{rate} = \{k_2 k_1 / k_{-1}\} [\text{NO}]^2 [\text{Br}_2]$$

$$k = k_2 k_1 / k_{-1}$$

Sample Exercise

Identify the catalyst in the reaction mechanism below:



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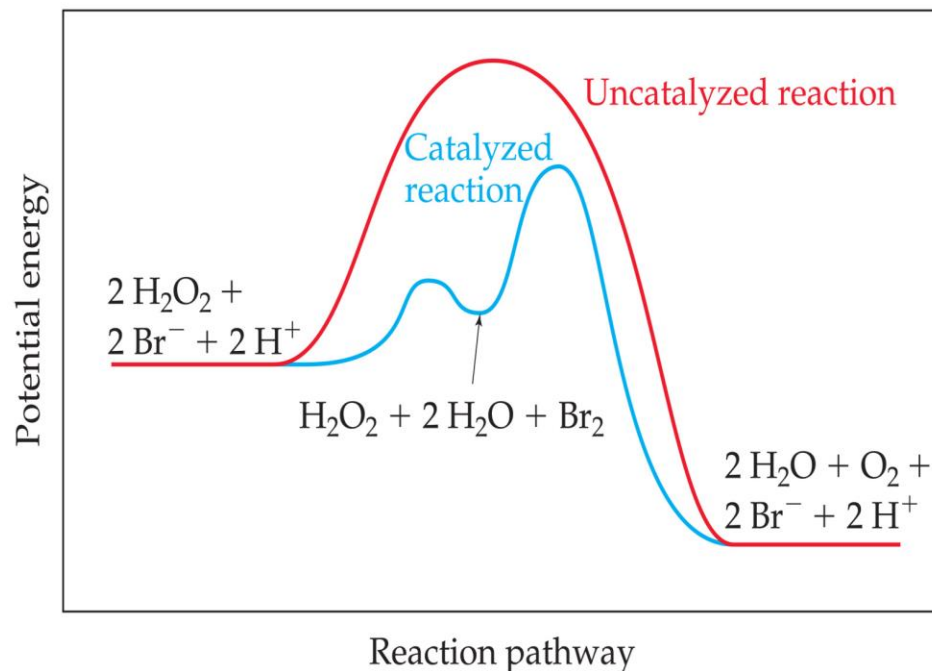
14.6 Catalysis

- A catalyst speeds up a reaction without being consumed.

$$E_{ac} < E_{auc}$$

$$\Delta E_{rxn} = E_{products} - E_{reactants}$$

unaffected by catalyst



- Provides a set of elementary steps with more favourable kinetics than those that exist in its absence
- Many times a catalyst lowers the activation energy
- Enhances the rate of the reverse and forward reaction equally.

Types of Catalysts

- **Heterogeneous catalysts:** reactants and catalyst are in different phases
- **Homogeneous catalysts:** reactants and catalysts are dispersed in single phase
- **Enzyme catalysts:** biological catalysts